Keywords: pretraining • reinforcement learning • scaling laws • chess reasoning • post-training

Understanding Reasoning from Pretraining to Post-Training

NYU and Collaborators Use Chess as a Controlled Testbed to Uncover a Joint Pretraining-to-RL Scaling Law and Show That RL Concentrates Rather Than Creates Competence

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arXiv:2607.16097 • July 17, 2026

Reinforcement learning post-training has become a core ingredient in every state-of-the-art language model, yet the field lacks basic answers to two questions: how do pretraining choices shape the returns to RL compute, and what does RL actually do to the model? A team from NYU, Modal Labs, UCLA, UIUC, and Columbia now answers both questions using chess — a fully observable reasoning domain where every move can be evaluated with ground-truth certainty.

AT A GLANCE

Problem: RL post-training is studied in isolation from pretraining, leaving the joint compute allocation problem and RL's mechanistic role undefined.

Why it matters: Understanding the pretraining-to-RL interface is necessary for principled scaling decisions and for knowing what RL can and cannot add to a model.

Method: Pretraining of 5M–1B parameter models on chess games, SFT on synthetic reasoning traces, RL on chess puzzles; move-level probability analysis across checkpoints.

Result: A joint scaling law $R(L_{\text{pre}}, C_{\text{RL}})$ fits held-out runs accurately; RL concentrates probability mass on high-value moves already present in the pretrained distribution.

Evidence: Scaling law extrapolates accurately to a 1B-parameter model with 10× more RL compute than any training run used to fit it.

The Big Picture

Reinforcement learning post-training is now applied to every frontier model, yet the field studies it in isolation: RL is treated as a plug-in module applied to an already-trained base model, with no principled account of how the two training stages interact. This isolation has produced two blind spots. First, practitioners cannot predict how much RL compute a given pretrained model will reward — leading to costly trial-and-error compute allocation. Second, there is no controlled mechanistic account of what RL actually modifies in the model's representations and output distributions; the prevalent intuition is that "RL teaches reasoning" but this is rarely examined at the level of individual predictions.

The paper uses chess as a controlled testbed because it offers ground truth for every decision, a known policy complexity profile, and a dataset of human games spanning all skill levels — ideal conditions for clean scaling experiments that would be impossible on open-ended language tasks.

Why Existing Approaches Fall Short

Prior work on RL for reasoning studies either the RL stage in isolation (holding pretraining fixed) or varies pretraining at large scale without controlling the RL stage. Neither approach can identify the joint interaction between the two stages.

Studying RL in isolation conflates the contribution of the pretrained representation with the contribution of RL optimization, making it impossible to determine whether a performance gain came from RL itself or from the quality of the starting point. Conversely, varying pretraining at fixed RL compute answers which base model RL benefits most, but cannot characterize the functional form of that benefit or reveal what RL does to the model's internal decision-making.

Core Mechanism

The authors vary both pretraining quality (model size and data volume) and RL compute in a factorial design, measuring post-RL reward for each combination. They fit a parametric scaling law to the resulting data and validate it on held-out (pretraining quality, RL compute) pairs. In parallel, they analyze the move-probability distributions of each checkpoint to track how RL reshapes the policy beyond aggregate reward.

Technical Formulation

The joint scaling law expresses post-RL reward R as a function of pretraining loss L_{pre} and RL compute C_{RL}:

$$R(L_{\text{pre}}, C_{\text{RL}}) = R_\infty - \phi(L_{\text{pre}}) \cdot C_{\text{RL}}^{-\gamma}$$

where R_∞ is the asymptotic reward ceiling, $\phi(L_{\text{pre}})$ is a monotonically increasing function of pretraining loss (worse pretraining shifts the ceiling down and slows RL convergence), and γ is a universal power-law exponent estimated from data.

The function ϕ is fit empirically and found to be approximately linear in L_{pre} over the studied range, giving:

$$\phi(L_{\text{pre}}) \approx \alpha \cdot L_{\text{pre}} + \beta$$

with α and β estimated jointly with γ via nonlinear least squares. This form implies that the marginal value of RL compute is linear in pretraining loss: every unit improvement in pretraining quality proportionally increases the RL compute budget required to reach any given reward threshold.

Learning Procedure

Training proceeds in three stages applied to each factorial combination:

1. **Pretraining:** Next-token prediction on Lichess PGN game records. Models range from 5M to 1B parameters; data volume is varied orthogonally from 10M to 2B tokens.
2. **Supervised fine-tuning:** Training on synthetically generated chain-of-thought traces that articulate candidate move rationale in natural language.
3. **RL post-training:** GRPO with binary correctness reward on chess puzzles; no intermediate-step reward. RL compute is varied from 10^{18} to 10^{20} FLOPs across factorial conditions.

Experimental Findings

The joint scaling law fits held-out (pretraining, RL compute) pairs with relative error below 3% across all studied configurations, confirming that the parametric form captures the essential interaction between the two training stages.

Condition	Predicted R	Observed R
1B params, low RL	0.81	0.80
1B params, high RL	0.89	0.91
70M params, high RL	0.74	0.73
5M params, high RL	0.51	0.52

Mechanistic analysis reveals that RL concentrates the probability distribution: the probability of the ground-truth top move increases by an average of 31% after RL for positions where the pretrained model already ranks it top-1, while the probability assigned to moves not in the pretrained model's top-10 changes by less than 0.3% on average.

Ablations and Interpretation

SFT vs. RL: SFT on reasoning traces reaches a lower performance ceiling than RL, suggesting that imitation of human reasoning traces is insufficient and RL's self-play signal is necessary to reach high puzzle accuracy.

RL without SFT: Applying RL directly to the pretrained model (without SFT) converges more slowly and to a lower final reward, confirming that the chain-of-thought format learned during SFT provides a useful scaffold for RL exploration.

Mechanistic result: RL does not inject knowledge of moves the pretrained model assigned near-zero probability to. This has a direct implication: RL post-training cannot recover from severe pretraining data gaps. A model that has never seen a particular chess strategy in pretraining will not learn it through RL, regardless of RL compute budget.

Scaling extrapolation: The scaling law predicts post-RL reward for a 1B-parameter model trained with 10× more RL compute than any run used to fit the law, with only 1.8% relative error, suggesting the law is reliable for planning compute allocation at larger scales.

Reference

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